PawConnect

An AI-Assisted Web Platform for Integrated Dog Health Monitoring, Veterinary Connectivity, and Stray dog Welfare Engagement

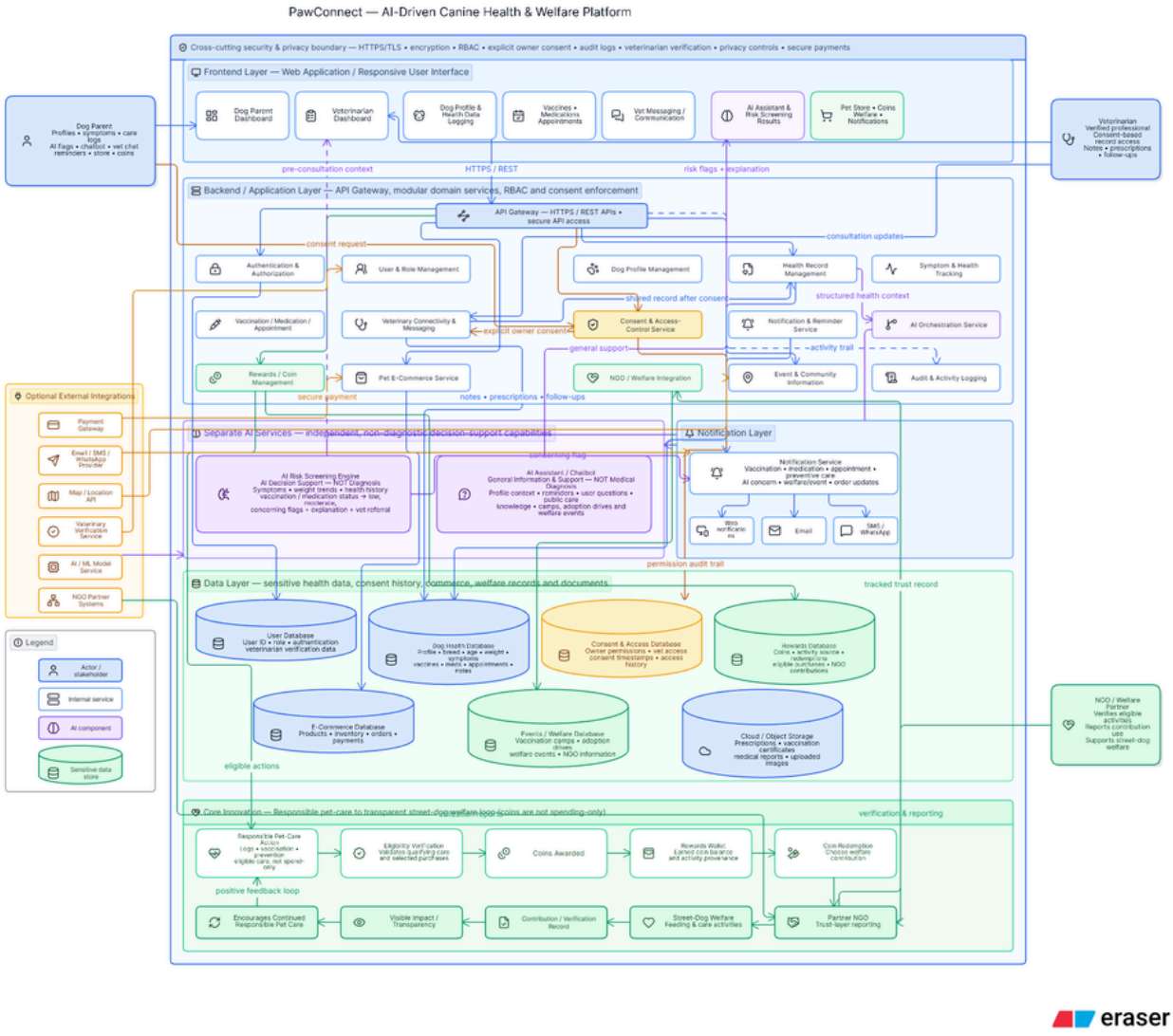
Abstract

Pet ownership has grown steadily, particularly among dog parents, yet the surrounding ecosystem of care remains fragmented across veterinary consultations, health record keeping, preventive care reminders, pet care applications, e-commerce platforms, and animal welfare initiatives. Dog parents often lack a centralized system to monitor their pet's health, retain medical history, receive timely reminders, and communicate structured information to veterinarians, while veterinarians frequently have limited visibility into a patient's condition between consultations. Existing solutions individually address parts of this ecosystem: telemedicine platforms enable remote consultation, record keeping applications store vaccination and medication data, e-commerce platforms sell pet products with loyalty incentives, and standalone charity platforms support stray animal welfare. However, no widely available system integrates these functions around a continuous, consent based owner and veterinarian relationship. This project proposes PawConnect, a web based, role based platform built using React.js and Spring Boot with Java, that connects dog parents and veterinarians through shared, structured health records, supported by a modular backend architecture comprising authentication and role management, dog profile and health record services, veterinary connectivity, and a dedicated AI orchestration layer. The platform incorporates an AI assisted health monitoring component that analyzes reported symptoms, weight trends, and health history to generate preliminary risk indications, categorized by concern level, intended strictly as decision support and early warning information rather than diagnosis, with veterinary professionals retaining responsibility for clinical judgement. A complementary AI assistant helps pet parents interpret recorded health information, delivers personalized vaccination, medication, and appointment reminders, and surfaces relevant pet festivals, adoption drives, vaccination camps, and animal welfare events to encourage community participation. All data access is governed by an explicit consent and access control service that logs every permission grant and record view, addressing the privacy and veterinary oversight concerns raised in the surrounding literature. The platform further introduces a reward mechanism in which pet parents earn coins through responsible pet care activities such as maintaining health records, completing preventive care tasks, and eligible purchases through an integrated pet store, with coins verified and redeemable through participating non-governmental organizations to support street dog feeding and welfare. While each underlying capability draws on established practice in veterinary telemedicine, pet health record keeping, and AI assisted screening, PawConnect's principal contribution lies in integrating continuous health monitoring, AI assisted guidance, and coin based animal welfare contribution within a single owner and veterinarian platform, promoting responsible pet ownership while extending measurable benefit to community animal welfare.

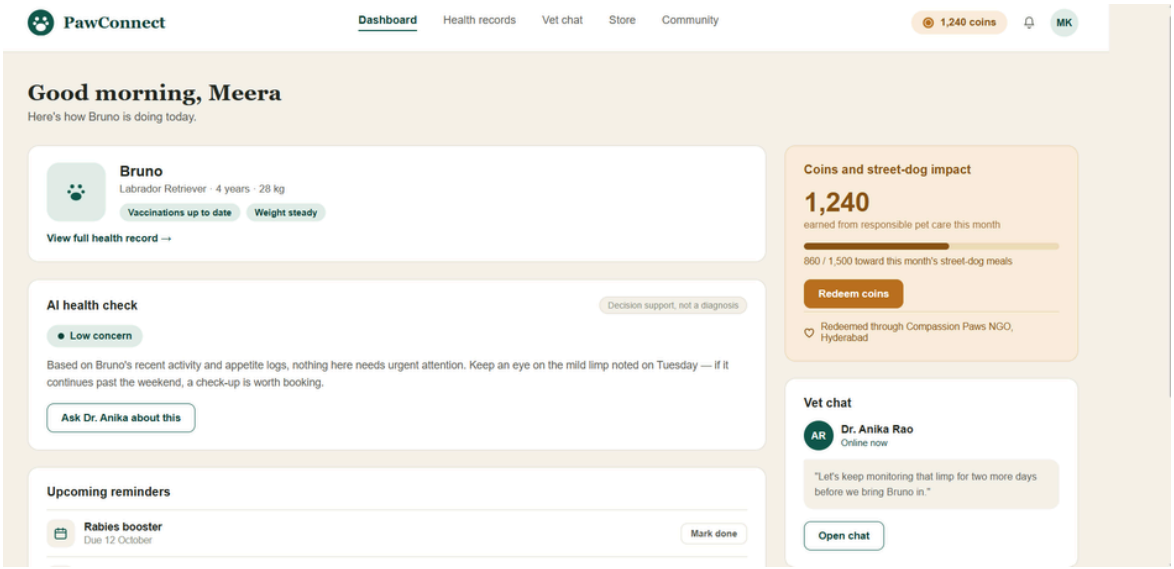
Keywords

Artificial Intelligence; Veterinary Informatics; Pet Health Monitoring; Machine Learning; Telemedicine; Animal Welfare; Preventive Care Reminders; Role-Based Access Control

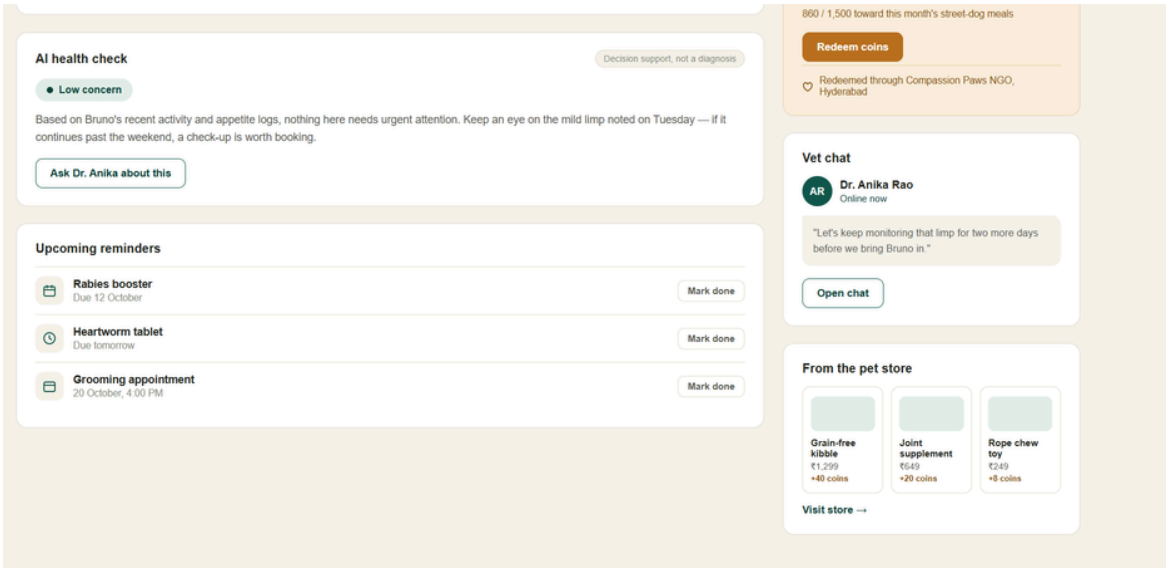
System Architecture



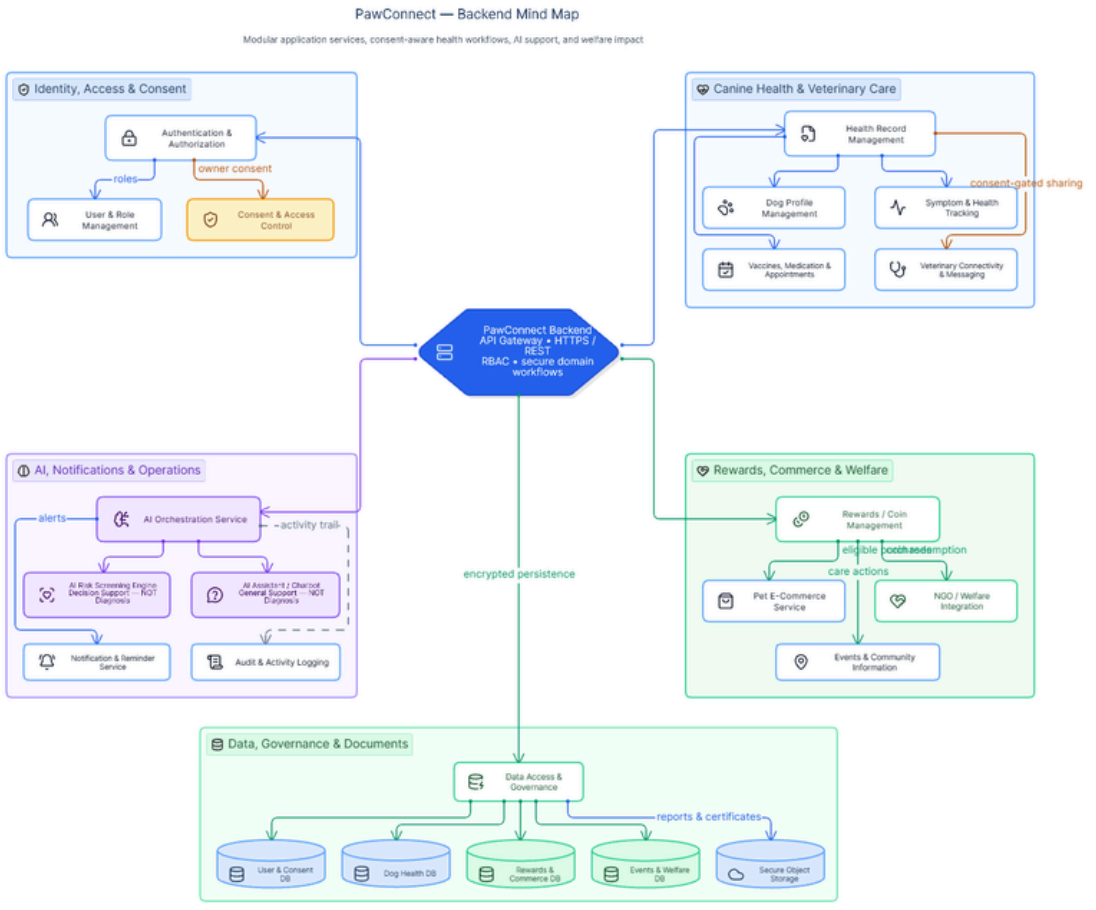
Example frontend prototype : pet parent dashboard



e-commerce platform and AI integrated health check up



Backend Mindmap



eraser